

Clinical Study Summary Report (CSSR)

Document Status: Draft | Version: 0.1 | Date: 03-Apr-2026

1. Administrative & Study Identification

Full Study Title: An Open-label Study to Assess the Safety, Efficacy, and Cellular Kinetics of YTB323 in Relapsing Multiple Sclerosis **Short Title/Acronym:** YTB323 in RMS **Protocol Number:** CYTB323N12101 **NCT Number:** NCT06617793 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** YTB323 (Rapcabtagene autoleucel; a CD19-directed CAR-T cell therapy) **Phase of Development:** Phase 1 / Phase 2 **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To assess the safety, tolerability, and cellular kinetics of ascending single doses of YTB323 in patients with Relapsing Multiple Sclerosis. **Secondary Objectives:** To evaluate the efficacy of YTB323 on disease activity (e.g., MRI lesion changes, Expanded Disability Status Scale [EDSS]), and to assess cognitive and motor function changes up to 2 years post-infusion.

2.2 Background & Rationale To address the unmet medical need in Relapsing Multiple Sclerosis (RMS) patients who experience breakthrough disease activity despite being treated with highly efficacious therapies (BD-HET). YTB323 is an autologous CAR-T cell therapy manufactured using T-Charge™ technology, which preserves T-cell "stemness" and targets CD19+ B-cells. This approach aims to tackle the immunopathological core of MS by effectively resetting the immune system and slowing disease progression.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Intervention Model:** Single Group Assignment (Ascending single-dose design with 3 sentinel cohorts followed by an expansion cohort) **Blinding:** None (Open-label) **Primary Purpose:** Treatment **Randomization:** N/A

3.2 Planned Enrollment & Duration **Target N:** 28 participants **Number of Sites:** Multi-center **Estimated Study Start:** 24-Feb-2025 **Estimated Primary Completion:** 01-Oct-2030 **Estimated Study Completion:** 01-Oct-2030

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 18 to 55 years old (Sex: All). 2. Confirmed diagnosis of Relapsing Multiple Sclerosis (RMS). 3. Evidence of breakthrough disease activity during previous treatment with a highly efficacious therapy (BD-HET).

4.2 Exclusion Criteria 1. Patients with non-active Progressive MS or other non-relapsing forms of MS (evaluated in separate protocols). 2. Lack of breakthrough disease activity on prior treatments. 3. Inability to safely undergo lymphodepleting chemotherapy or significant underlying medical conditions contraindicating CAR-T therapy.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single dose administration of YTB323 (Rapcabtagene autoleucel) across different ascending dose cohorts. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Single infusion with a primary outcome tracking and follow-up period of up to 2 years.

5.2 Concomitant & Prohibited Therapy Permitted: Pre-treatment lymphodepleting chemotherapy using intravenous cyclophosphamide and fludarabine phosphate. Standard management medications for adverse events, such as tocilizumab, for the management of potential Cytokine Release Syndrome (CRS). **Prohibited:** Other systemic disease-modifying therapies (DMTs) for MS during the active study period to avoid confounding efficacy and safety data.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to Day -7	Informed Consent, Medical History, Baseline MRI, EDSS, Cognitive tests
Baseline	Day -6 to Day 0	Lymphodepletion (Cyclophosphamide + Fludarabine), Baseline Labs
Interim	Day 1 to Month 24	YTB323 IV Infusion, CRS/Neurotoxicity monitoring, Cellular Kinetics (PK blood draws), Efficacy Assessments (MRI, Timed 25-Foot Walk, 9-Hole Peg Test, SDMT)
End of Study	Year 2 (Week 104)	Final EDSS, Final MRI scans, Exit Interview, Transition to long-term safety follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence of dose-limiting toxicities (DLTs), Adverse Events (AEs), and Serious Adverse Events (SAEs) up to 2 years. Cellular kinetics including Maximum

plasma concentration ($C_{\max}$) and Area Under the Curve (AUC) of the drug up to 2 years after infusion. **Measurement Method:** Clinical assessment, vital signs, ECGs, blood sample assays, and brain/spinal cord MRIs.

7.2 Secondary & Exploratory Endpoints * Measure of Disability: Change in Expanded Disability Status Scale (EDSS) score (0 to 10 scale). * Number of new and enlarging T2 lesions and Gd-enhancing T1 lesions on MRI. * Motor and cognitive function: Timed 25-Foot Walk (T25FW), 9-Hole Peg Test (9HPT), and Symbol Digit Modalities Test (SDMT). * Patient-Reported Outcomes: Fatigue Symptoms and Impacts Questionnaire (FSIQ-RMS) and Short-form health survey for quality of life.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Specialized monitoring for CAR-T associated risks, including Cytokine Release Syndrome (CRS) and neurotoxicity. **Serious Adverse Events (SAEs):** Standard 24-hour reporting timeline. Intensive tracking post-infusion for neurological status and immunological responses. **Data Monitoring Committee (DMC):** Internal and external safety review of the 3 sentinel cohorts prior to progressing to the expansion cohort.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients receiving lymphodepletion or YTB323); Efficacy Analysis Set (Intent-to-Treat for efficacy parameters). **Sample Size Justification:** $N = 28$, an appropriate size for an ascending single-dose phase 1/2 trial designed to evaluate DLTs, safety, and initial cellular kinetics across cohorts. **Handling of Missing Data:** Standard phase 1/2 handling utilizing mixed-effects models for repeated measures (MMRM) or Last Observation Carried Forward (LOCF) for longitudinal disability and cognitive scoring.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Intensive onsite monitoring post-infusion followed by remote monitoring and long-term registry tracking for cellular therapy products. **Audit Trail:** Strict eCRF locking, tracking of CAR-T manufacturing (vein-to-vein time), and release specifications verification.

11. Study Identifiers & Governance

Primary ID: CYTB323N12101 **Registry Number:** NCT06617793 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** An Open-label Study to Assess the Safety, Efficacy, and Cellular Kinetics of YTB323 in Relapsing Multiple Sclerosis **Official Regulatory Title:** An Open-label, Multi-center, Non-confirmatory Study to Assess the Safety, Efficacy, and Cellular Kinetics of YTB323 in Relapsing Multiple Sclerosis

12. Clinical Framework (Multiple Sclerosis Specific)

Primary Condition: Relapsing Multiple Sclerosis (RMS) **Diagnosis Threshold:** Breakthrough disease activity during previous treatment with highly efficacious therapy (BD-HET) **Medical Methodology:** Autologous CD19-directed CAR-T cell therapy **Optimization Target:** Depletion of disease-driving B cells to reset the immune system

13. Study Procedures & Methodology

Management Pathway: Leukapheresis → YTB323 manufacturing → Lymphodepleting chemotherapy → Single IV Infusion of YTB323. **Education Protocol (HCP):** Specialized training on the recognition and management of Cytokine Release Syndrome (CRS) and neurotoxicity specific to CAR-T cell therapies. **Education Protocol (Patient):** Education on the risks of immunosuppression, CRS symptoms, and the requirement for long-term safety follow-ups. **Quality Audit Mechanism:** Cellular kinetics profiling and humoral/cellular immunogenicity tracking.

14. Observation & Follow-Up Schedule

Total Duration: 24 Months (2 years) post-infusion **Onsite Visit Cadence:** Leukapheresis, Infusion Day, and intense follow-up schedule typically spanning Days 1-28, then Months 3, 6, 12, 18, and 24. **Remote Monitoring Frequency:** Routine safety check-ins post-discharge. **Checklist Review Interval:** Cognitive exams (SDMT), physical assessments (EDSS, T25FW, 9HPT), and MRI scans conducted sequentially up to 2 years.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				